Probability Theory - Winter term 2025/26
Personal Solution Set 5

TU Dortmund

Problem 1

Let $f_n(x) = n\sin(nx)$ for $0 \leq x \leq \pi/n$, and 0 otherwise.

i) Show that f_n is integrable for all $n \in \mathbb{N}$.

ii) Show that f_n converges pointwise.

iii) Does f_n fulfill the requirements for Fatou's Lemma and the DCT?

iv) Does $\lim_{n\to\infty} \int f_n d\mu = \int \lim_{n\to\infty} f_n d\mu$ hold?

Solution

i) The function is non-negative on $[0, \pi]$. I compute the Lebesgue integral as a Riemann integral:
$$\int_0^\pi |f_n(x)|dx = \int_0^{\pi/n} n\sin(nx)dx = [-\cos(nx)]_0^{\pi/n} = -\cos(\pi) + \cos(0) = 2.$$
Since $2 < \infty$, f_n is integrable for all n.

ii) For $x = 0$, $f_n(0) = n\sin(0) = 0 \to 0$. For any fixed $x \in (0, \pi]$, there exists an n_0 large enough such that for all $n \geq n_0$, $x > \pi/n$. For these n, $f_n(x) = 0$. Thus, $f_n(x)$ converges pointwise to $f(x) = 0$ everywhere on $[0, \pi]$.

iii) Fatou's Lemma applies because $f_n \geq 0$ everywhere and converges pointwise. The Dominated Convergence Theorem (DCT) does not apply. I cannot find an integrable bounding function $g(x)$ such that $|f_n(x)| \leq g(x)$ for all n. As n increases, the peak of f_n reaches n, which diverges to infinity.

iv) The equality does not hold. From item (i), $\lim_{n\to\infty} \int f_n d\mu = \lim_{n\to\infty} 2 = 2$. However, the integral of the limit is $\int 0 d\mu = 0$. Since $2 \neq 0$, the limits cannot be swapped.

Problem 2

Let $X_n = n1_{[0,1/n]}$ on $([0, 1], \mathcal{B}([0, 1]), \lambda_{[0,1]})$.

i) Show $X_n \xrightarrow{p} 0$.

ii) Does $X_n \xrightarrow{a.s.} 0$?

iii) Calculate limits of expectation and expectation of limit.

iv) Does X_n fulfill MCT requirements?

v) Does X_n fulfill DCT requirements?

Solution

i) Let $\epsilon > 0$. For $n > \epsilon$, the probability is exactly the measure of the support interval:

$$P(X_n > \epsilon) = \lambda\left(\left[0, \frac{1}{n}\right]\right) = \frac{1}{n} \xrightarrow{n\to\infty} 0.$$

Thus, X_n converges in probability to 0.

ii) Yes. For any $\omega \in (0, 1]$, $X_n(\omega) = 0$ for all $n > 1/\omega$. The only point where convergence fails is $\omega = 0$, where $X_n(0) = n \to \infty$. However, $\lambda(\{0\}) = 0$. The sequence converges pointwise to 0 almost everywhere.

iii) The expected value of the pointwise limit is $E(\lim X_n) = \int 0\, d\lambda = 0$. The limit of the expectations is $\lim_{n\to\infty} E(X_n) = \lim_{n\to\infty} \left(n \cdot \frac{1}{n}\right) = 1$. The equality does not hold.

iv) The MCT requires a monotone increasing sequence ($X_n \leq X_{n+1}$). This sequence is not increasing; for example, $X_1(1) = 1$, but $X_2(1) = 0$.

v) The DCT requires a uniform integrable bound C. Just like Problem 1, the peak value reaches n, meaning no finite bounding function exists. The DCT cannot be applied.

Problem 3

Evaluate $\lim \int f_n d\lambda$ and $\int \lim f_n d\lambda$ for two sequences.

i) $f_n(x) = nx^n 1_{[0,1)}(x)$.

ii) $f_n(x) = \sum_{i=1}^{n} i 1_{[i,i+1)}(x) - a 1_{[n,n+1)}(x)$ with $a > 1$.

Solution

i) The pointwise limit for $x \in [0, 1)$ is $\lim_{n\to\infty} nx^n = 0$. Thus, $\int \lim f_n d\lambda = 0$. The limit of the integral is:

$$\lim_{n\to\infty} \int_0^1 nx^n dx = \lim_{n\to\infty} \left[\frac{n}{n+1} x^{n+1}\right]_0^1 = \lim_{n\to\infty} \frac{n}{n+1} = 1.$$

The limits are not equal. MCT fails because the sequence is not monotonically increasing, and DCT fails because there is no integrable bound.

ii) The pointwise limit is $\lim f_n(x) = \sum_{i=1}^{\infty} i 1_{[i,i+1)}(x)$ because the subtracted term vanishes as $n \to \infty$. The integral of this limit diverges: $\int \lim f_n d\lambda = \sum_{i=1}^{\infty} i = \infty$. The limit of the integral is:

$$\lim_{n\to\infty} \int_{\mathbb{R}} f_n d\lambda = \lim_{n\to\infty} \left(\sum_{i=1}^{n} i - a\right) = \lim_{n\to\infty} \left(\frac{n(n+1)}{2} - a\right) = \infty.$$

They evaluate to the same infinity. MCT fails because the sequence isn't strictly monotone (depending on a), and DCT fails because the bounding function would need to encompass the entire infinite series, which is not integrable.